

Sahana A

Senior Data Engineer

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SUMMARY

- 9+ years of experience in designing and managing end-to-end **ETL/ELT pipelines**, data ingestion frameworks, and **enterprise data warehouse** solutions on **Microsoft Azure**.
- Built scalable data workflows using **Azure Data Factory**, **Synapse Analytics**, **Microsoft Fabric**, **Azure Data Lake**, and **Azure Databricks** for batch and real-time processing.
- Developed streaming data pipelines using **Apache Kafka**, **Azure Event Hubs**, and **Spark Streaming**, incorporating **Change Data Capture (CDC)** to support real-time analytics.
- Implemented **data transformation and modeling** using **SQL**, **Python**, **PySpark**, and **DBT**, leveraging **Spark SQL**, **Pandas**, **NumPy**, and **SciPy** for analytics and ML use cases.
- Experienced using **Azure Resource Manager (ARM)** templates to automate infrastructure deployment, ensuring consistency and efficiency.
- Developed an application using **Spark** for data processing and analysis of customer data.
- Experience in implementing **Spark** using **Scala** and **Spark SQL** for faster processing of data.
- Engineered **data warehouses** in **Snowflake** and **Azure Synapse Analytics**, using **partitioning**, **clustering**, and **indexing** strategies to optimize performance and cost-efficiency.
- Good understanding of different file formats like **JSON**, **Parquet**, **Avro**, **ORC**, **Sequence**, **XML** etc.
- Experience in collecting and storing stream data like log data in HDFS using **Flume**.
- Experience in performing data validation using **HIVE dynamic partitioning and bucketing**.
- Involved in creating tables, partitioning, bucketing of tables and creating UDF's in **Hive**.
- Experienced in job workflow scheduling and monitoring tools like **Oozie** and **Zookeeper**.
- Worked on importing and exporting data from different databases like Oracle, MySQL into HDFS and Hive using **Sqoop**.
- Applied security best practices using **Azure Active Directory**, **Role-Based Access Control (RBAC)**, **Key Vault**, and **network security policies**, ensuring compliance with **GDPR**, **HIPAA**, and **PCI-DSS**.
- Automated data workflows and orchestrated complex pipeline dependencies using **Apache Airflow**, **NiFi**, and **DBT Cloud**, enabling end-to-end observability and reliability.
- Built CI/CD pipelines using **Azure DevOps**, **GitHub Actions**, **GitLab CI/CD**, and **Jenkins**, deploying containerized ETL services with **Docker** and managing infrastructure via **Kubernetes**.
- Provided mentorship and technical leadership by guiding junior engineers through code reviews, architectural planning, and hands-on sessions on **Databricks**, **Azure services**, and **Spark**.
- Delivered insightful dashboards and business reports through **Power BI** and **Tableau**, transforming raw data into actionable intelligence for decision-making.

SKILLS

Programming languages	Python, Shell Scripting, Perl Scripting, SQL, PL/SQL, Hive QL
Big Data Distributions, Reporting Tools & Eco Systems	Cloudera, Hortonworks, Apache Hadoop, HDFS, Map Reduce, YARN, Spark, PySpark, Power BI, Tableau
Scheduling & Monitoring Tools	Control-M, Cloudera Manager, Apache Airflow
Azure Stack	Azure Blob Storage, Azure Data Lake Storage (ADLS Gen2), Azure Databricks, Azure Data Factory (ADF), Analytics, Azure Key-Vault, Azure Functions, Azure Cosmos DB, Azure DevOps, Azure Logic Apps, Azure HDInsight
Cloud Platforms	Azure, Databricks, AWS (S3, EC2, IAM, Lambda, Redshift)
Messaging Queue/ Data Services	Hive, Sqoop, Flume, Kafka

Relational & No-SQL Databases	Oracle, MySQL, MS SQL, Cassandra, HBase, MongoDB.
ETL/ELT & Data Engineering	Azure Data Factory (ADF), Synapse Pipelines, CDC tools, PolyBase, Data Modeling (Star Schema, Snowflake Schema), Delta Lake, Data Lakes
File Formats	ORC, AVRO, Parquet, CSV, JSON, XML, Delta File Formats.
IDE / Developmental Tools	PyCharm, Jupyter Notebook, Spyder, Visual Studio, Google Colab, Eclipse, STS, Swagger, SQL YOG, MySQL Workbench, Sonar Cube, Presto, Snowflake Workspace
Other Software Tools	ELK Stack, Postman, TOAD, SQL developer tool, Jira.
Version Control & Build Tools	CVS, SVN & Bit Bucket (GIT), Maven, Jenkins (DevOps), Ant, Gradle

EXPERIENCE

Senior Azure Data Engineer | Walgreens | May 2023 - Present

- Built ingestion workflows with Informatica and Azure Data Factory to load structured datasets from internal and external sources into Data Lake and Synapse for downstream processing.
- Developed ETL scripts in Python on Databricks and Azure Functions to perform custom data transformations, improving workflow flexibility.
- Automated workflows for electronic health records (EHR) and patient demographics using Python, SQL, and Azure Data Factory, improving data availability by 35%.
- Migrate existing ETL (Extract, Transform, Load) processes from Teradata to Azure, considering the capabilities of Azure services such as Azure Data Factory.
- Developed and optimized SQL queries in Synapse to support dashboards, reporting, and ad hoc analysis across large datasets.
- Improved PySpark jobs by using partitioning, caching, and broadcast methods to process large datasets more effectively, making queries faster and reducing overall execution time across pipelines.
- Built Spark applications using DataFrame API, RDD, and Spark SQL on Databricks to process batch and micro-batch workloads for healthcare and operational data.
- Reviewed and optimized existing Azure Data Factory pipelines and Synapse queries to achieve better performance, resulting in 40% faster data delivery and 25% lower computing expenses.
- Integrated Delta Lake to support ACID transactions and improve data reliability.
- Troubleshoot and resolved performance issues in production Spark jobs and Airflow DAGs running on elastic Kubernetes environments, resulting in a 30% decrease in job failures.
- Implemented Azure Active Directory, Role-Based Access Control, and Key Vault to secure PHI through encryption, access control, and audit logging, applying HL7 FHIR standards to ensure secure and interoperable healthcare data exchange while maintaining compliance with HIPAA, GDPR, PCI-DSS, and ISO 27001.
- Designed CI/CD pipelines using Azure DevOps Pipelines and Terraform for infrastructure as code (IaC) deployment and automated testing of data applications within containerized environments managed by Docker and Azure Kubernetes Service.
- Created Power BI dashboards which included custom visuals and audit tracking features to fulfill compliance needs and deliver business intelligence for reporting purposes.

Environment: Azure Data Factory, Azure SQL Database, Azure Data Lake, Azure Blob Storage, Azure Synapse Analytics, Azure Functions, Microsoft Fabric, Azure Cosmos DB, Azure Event Hubs, Azure Active Directory, Azure Key Vault, Azure Kubernetes Service, Azure DevOps, Informatica, Python, SQL, Pandas, PySpark, Scikit Learn, RDD, Spark SQL, Spark Core, DataFrame API, EHR, IaC, Databricks, Delta Lake

ETL Developer / Data Stage Developer | Infosys | Mar 2011 – July 2017

- Designed, developed, and maintained end-to-end ETL pipelines using IBM DataStage and Informatica, ensuring accurate and timely data delivery.
- Built and optimized SQL, PL/SQL, and T-SQL queries and stored procedures for complex transformations and large-volume data loads.
- Implemented parallel DataStage jobs and performed debugging using Peek, Row Generator, Head & Tail, and Column Generator stages.
- Developed UNIX (ksh) shell scripts for pre-load/post-load validations, auditing, file handling, and automated job notifications.
- Scheduled and monitored batch jobs using Control-M, ensuring SLA adherence and resolving production incidents and change requests.
- Created low-level and high-level design documents, unit test plans, and transformation logic documentation.
- Optimized ETL performance by separating extraction, transformation, and load processes, reducing batch load windows.
- Utilized PX stages (Sequential File, DataSet, Lookup, Complex Flat File) for scalable data processing.
- Supported analytics and reporting by delivering clean, validated datasets for Tableau and SAS consumption.
- Developed Hive tables and implemented partitioning, dynamic partitions, and bucketing for performance optimization.
- Built Oozie workflows and Sqoop jobs to ingest data from Teradata into Hive (ORC/Parquet formats).
- Managed Hadoop ecosystem components including HDFS, Hive, logs, and batch jobs in a distributed environment.
- Automated multi-source data integration workflows, reducing manual effort and improving data availability by ~30%.
- Collaborated with business analysts and QA teams to define data quality rules and validation checks.
- Mentored junior team members on SQL optimization, ETL best practices, and testing strategies.

Environment: SQL, PL/SQL, UNIX (ksh), IBM DataStage, Informatica, Control-M, Hadoop, HDFS, Hive, Sqoop, Oozie, Databricks, DBFS, Parquet, Data Lake, Jenkins, GitHub, DB2, SAS, Tableau, Flat Files, Batch Processing, Data Pipelines, Data Validation, Regulatory Reporting

EDUCATION

Bachelors in Electronics and Communication Engineering – Visveswaraya Technological University

CERTIFICATIONS

- Microsoft Certified: Azure Developer Associate
- Databricks Certified: Data Engineer Associate
- IBM Certified Solution Developer InfoSphere DataStage v8.0 Infosys 2013-03-03
- IBM Certified Solution Developer InfoSphere DataStage v8.05 Infosys 2015-06-05